

Reciprocal Rational Maps, Self-Dual Hinges, and Toroidal Phase Lifts

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Abstract

We study a family of reciprocal rational maps

$$f(x) = \frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A},$$

which possess a fundamental inversion symmetry

$$f(x)f(1/x) = 1.$$

By introducing the folded coordinate

$$t = x^2 + \frac{1}{x^2},$$

the dynamics reduce to algebraic shell structures governed by quadratic lock equations. This reduction reveals a distinguished self-dual hinge at $x = 1$ and reciprocal shell classes that emerge from the real solutions of the folded system.

We analyze parameter deformations of the map and identify regimes characterized by a single reciprocal shell, multiple shells, or hinge-dominated configurations. A toroidal phase lift is then introduced, providing a geometric carrier for phase transport that enforces cyclic traversal of shell states.

The resulting framework combines algebraic reciprocity, folded dynamical systems, and toroidal phase topology into a unified geometric interpretation.

1 Introduction

Reciprocal structures appear in many areas of mathematics and physics, particularly in systems where inversion symmetries constrain the dynamics.

In this work we investigate a rational dynamical system with built-in reciprocal symmetry:

$$f(x) = \frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A}.$$

This map exhibits a natural inversion invariance:

$$f(x)f(1/x) = 1.$$

Such symmetry suggests that the dynamics should naturally organize into reciprocal shell structures around a central self-dual point.

We show that the map admits a natural folded coordinate

$$t = x^2 + \frac{1}{x^2},$$

which reduces the lock conditions of the system to quadratic equations. This reveals a simple algebraic structure underlying the dynamics.

2 The Reciprocal Rational Map

Consider the family of maps

$$f(x) = \frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A}.$$

A direct computation shows

$$f(1/x) = \frac{1}{f(x)}.$$

Thus

$$f(x)f(1/x) = 1.$$

This property implies that the dynamics respect inversion symmetry $x \leftrightarrow 1/x$.

3 Reciprocal Folding

Define the folded coordinate

$$t = x^2 + \frac{1}{x^2}.$$

The quantity t is invariant under inversion:

$$x \rightarrow \frac{1}{x}.$$

Thus the reciprocal dynamics can be analyzed on the folded variable t .

This transformation converts the lock equations of the system into quadratic relations.

4 Lock Structure

Two types of locks arise naturally.

4.1 Fixed Locks

Fixed locks satisfy

$$f(x) = x.$$

In the folded coordinate this reduces to

$$t^2 - (A^2 - 2B - 1)t - (A^2 + 2AB - 2A - B^2 - 2B + 1) = 0.$$

4.2 Reciprocal Locks

Reciprocal locks satisfy

$$f(x) = \frac{1}{x}.$$

The folded equation becomes

$$A^2 t^2 + (A^2 + 2AB - 1)t + (-A^2 + 2AB + 2A + B^2 - 2B - 1) = 0.$$

Real shell states correspond to solutions satisfying

$$t \geq 2.$$

5 The Self-Dual Hinge

A distinguished point occurs at

$$x = 1.$$

This point satisfies both lock conditions:

$$f(1) = 1, \quad f(1) = \frac{1}{1}.$$

Thus $x = 1$ acts as a **self-dual hinge** in the dynamical structure.

This hinge forms the algebraic center of the reciprocal shell geometry.

Along the diagonal family $A = B = \lambda$, the reciprocal-lock sector collapses below the real-shell threshold, leaving the hinge and fixed-lock shell structure as the dominant real scaffold.

Theorem 1 (Self-Dual Hinge)

Theorem. For the reciprocal rational map

$$f(x) = \frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A},$$

the point $x = 1$ is simultaneously a fixed lock and a reciprocal lock. In particular,

$$f(1) = 1,$$

and hence $x = 1$ is a self-dual hinge of the dynamics.

Proof. Evaluating the map at $x = 1$ gives

$$f(1) = \frac{A + B + 1}{1 + B + A} = 1.$$

Thus $x = 1$ is a fixed point.

Since

$$\frac{1}{1} = 1,$$

the same point also satisfies the reciprocal lock condition

$$f(x) = \frac{1}{x}$$

when $x = 1$. Therefore $x = 1$ lies in both the fixed-lock and reciprocal-lock sets.

Hence $x = 1$ is a self-dual hinge for the reciprocal rational map. $\square$

The significance of Theorem 1 is structural rather than merely algebraic. The point $x = 1$ is the unique location at which self-identity and reciprocal identity coincide. For this reason it serves as the natural center of the folded shell geometry and provides the anchor point for the geometric lift introduced later in the paper.

Theorem 2 (Diagonal Reciprocal-Lock Reduction)

Theorem. Along the diagonal parameter family

$$A = B = \lambda,$$

the reciprocal-lock equation in the folded variable

$$t = x^2 + \frac{1}{x^2}$$

reduces to

$$\lambda^2 t^2 + (3\lambda^2 - 1)t + (2\lambda^2 - 1) = 0.$$

Its discriminant is

$$\Delta_r = (\lambda^2 - 1)^2,$$

and the corresponding folded roots are

$$t = -1, \quad t = -2 + \frac{1}{\lambda^2}.$$

For every $\lambda > 0$, both roots satisfy $t < 2$, and therefore do not produce additional real shell states in x -space.

Proof. Starting from the reciprocal-lock equation

$$A^2 t^2 + (A^2 + 2AB - 1)t + (-A^2 + 2AB + 2A + B^2 - 2B - 1) = 0,$$

set $A = B = \lambda$. This gives

$$\lambda^2 t^2 + (\lambda^2 + 2\lambda^2 - 1)t + (-\lambda^2 + 2\lambda^2 + 2\lambda + \lambda^2 - 2\lambda - 1) = 0,$$

which simplifies to

$$\lambda^2 t^2 + (3\lambda^2 - 1)t + (2\lambda^2 - 1) = 0.$$

Its discriminant is

$$\Delta_r = (3\lambda^2 - 1)^2 - 4\lambda^2(2\lambda^2 - 1) = 9\lambda^4 - 6\lambda^2 + 1 - 8\lambda^4 + 4\lambda^2 = \lambda^4 - 2\lambda^2 + 1 = (\lambda^2 - 1)^2.$$

Thus the roots are

$$t = \frac{-(3\lambda^2 - 1) \pm (\lambda^2 - 1)}{2\lambda^2}.$$

This yields

$$t = -1, \quad t = -2 + \frac{1}{\lambda^2}.$$

For every $\lambda > 0$, both values are strictly less than 2. Since real shell states require

$$t \geq 2,$$

these reciprocal-lock roots do not produce additional real shell branches. $\square$

Theorem 2 explains why the diagonal family $A = B = \lambda$ is structurally simple. Although reciprocal-lock solutions exist algebraically in the folded coordinate, they remain outside the real-shell threshold $t \geq 2$. Thus, along the positive diagonal family, the real shell scaffold is controlled by the hinge and the fixed-lock sector, rather than by the reciprocal-lock sector. This is the main reason the regime near $(A, B) = (9, 9)$ exhibits a clean single-shell structure.

Proposition 1 (Diagonal Shell Scaling)

Proposition. Along the diagonal parameter family

$$A = B = \lambda,$$

the real shell representatives of the reciprocal rational map satisfy the asymptotic scaling

$$|x| \sim \lambda, \quad |x| \sim \frac{1}{\lambda}.$$

Proof. From the fixed-lock equation in the folded coordinate

$$t = x^2 + \frac{1}{x^2},$$

the dominant folded root along the diagonal family behaves asymptotically as

$$t_+ \sim \lambda^2$$

for large λ .

Substituting into

$$x^2 + \frac{1}{x^2} = t_+,$$

and assuming $|x| > 1$, the dominant term is x^2 , giving

$$x^2 \sim t_+.$$

Thus

$$|x| \sim \sqrt{t_+} \sim \lambda.$$

Since the system is reciprocal under the transformation

$$x \rightarrow \frac{1}{x},$$

the second shell representative satisfies

$$|x| \sim \frac{1}{\lambda}.$$

This establishes the reciprocal scaling of the shell pair along the diagonal parameter family. $\square$

This proposition provides a geometric interpretation of the diagonal parameter family. The shell representatives expand and contract reciprocally with the parameter λ , while the hinge at $x = 1$ remains fixed. Consequently the dynamical scaffold consists of a stable central hinge surrounded by a reciprocal shell pair whose radius grows and shrinks smoothly with the system scale.

6 Shell Regimes

Theorems 1 and 2 identify two persistent features of the reciprocal rational map: the universal self-dual hinge at $x = 1$, and the suppression of additional real reciprocal-lock shells along the diagonal family $A = B = \lambda$. The remaining shell structure is therefore determined by the real folded roots of the fixed-lock sector.

Solutions of the folded lock equations determine the shell structure.

We classify parameter regimes as follows.

- H1S1 : hinge with one reciprocal shell
- H1S2 : hinge with multiple shells
- H1S0 : hinge only
- Broken : degenerate configurations

Numerical exploration indicates that $(A, B) = (9, 9)$ lies within the H1S1 regime.

7 Computational Exploration Method

The analytical structure of the reciprocal rational map was investigated using numerical exploration of the lock equations derived in the folded coordinate

$$t = x^2 + \frac{1}{x^2}.$$

For each parameter pair (A, B) , the system is analyzed through the following procedure:

1. Solve the fixed-lock quadratic

$$t^2 - (A^2 - 2B - 1)t - (A^2 + 2AB - 2A - B^2 - 2B + 1) = 0.$$

2. Solve the reciprocal-lock quadratic

$$A^2 t^2 + (A^2 + 2AB - 1)t + (-A^2 + 2AB + 2A + B^2 - 2B - 1) = 0.$$

3. Retain only solutions satisfying

$$t \geq 2,$$

which correspond to real values of x .

4. Recover the corresponding shell representatives using

$$z + \frac{1}{z} = t,$$

which yields

$$z = \frac{t \pm \sqrt{t^2 - 4}}{2}.$$

5. Lift these solutions to real shell coordinates

$$x = \pm\sqrt{z}.$$

6. Group reciprocal duplicates $(x, 1/x)$ into shell classes.

This procedure determines the real shell scaffold of the system for any parameter pair (A, B) .

The computational exploration reveals that near $(A, B) = (9, 9)$ the system exhibits a single reciprocal shell class surrounding the hinge $x = 1$, corresponding to the regime labeled H1S1 in the shell classification.

8 Visualization Framework

To better understand the structure of the reciprocal shell system, a three-dimensional visualization framework was developed.

The visualization maps algebraic shell representatives to geometric structures using the folded shell depth

$$D(x) = |\log_2 |x||.$$

Using Proposition 1, the shell depth grows approximately as

$$D(x) \sim |\log_2 \lambda|,$$

along the diagonal family $A = B = \lambda$. Thus the visualization represents parameter deformation as a smooth expansion or contraction of the reciprocal shell band around the hinge core.

This transformation identifies reciprocal values $(x, 1/x)$ as belonging to the same geometric shell level.

The resulting structure consists of the following components:

- **Hinge Core**

The self-dual hinge at $x = 1$ is represented as a central node from which shell connections emerge.

- **Reciprocal Shell Band**

Shell representatives derived from the lock equations are mapped to a common radial band using the folded shell depth $D(x)$.

- **Phase Replication**

A twelve-phase ring replicates shell states around the hinge, providing a phase index for the reciprocal shell structure.

- **Toroidal Carrier**

The phase structure is embedded on a toroidal manifold, which provides a natural topology for cyclic phase transport.

This geometric lift converts the algebraic shell scaffold of the reciprocal rational map into an interpretable spatial structure. The visualization does not introduce new dynamics but rather provides an explicit representation of the reciprocal hinge and shell relationships determined by the folded lock equations.

9 Parameter Phase Diagram

The reciprocal rational map exhibits different shell structures depending on the parameters (A, B) .

Using the computational exploration procedure described earlier, the system can be classified according to the number of real shell classes present in the folded coordinate.

We define the following structural regimes:

- **H1S1** : hinge with one reciprocal shell
- **H1S2+** : hinge with two or more shells
- **H1S0** : hinge only
- **Broken** : degenerate or numerically unstable configurations

Empirical exploration shows that the parameter pair

$$(A, B) = (9, 9)$$

lies within the H1S1 regime.

In this regime the system exhibits:

- a stable hinge at $x = 1$
- a single reciprocal shell class
- smooth shell deformation under small parameter changes

Along the diagonal family

$$A = B = \lambda$$

the reciprocal-lock sector remains below the real-shell threshold $t \geq 2$, leaving the hinge and fixed-lock shell structure as the dominant real scaffold.

Consequently the diagonal family forms a stable corridor of single-shell behavior in parameter space.

10 Toroidal Phase Lift

To incorporate phase transport, we embed the shell structure onto a toroidal domain

$$T_\pi = \mathbb{R}/\pi\mathbb{Z}.$$

A twelve-phase modulation produces the rosette structure

$$\cos(12x).$$

On the π -toroid the modulation integrates to zero:

$$\int_0^\pi \cos(12x)dx = 0.$$

Thus the topology enforces complete phase traversal across shell states.

11 Geometric Lift

The combined structure can be interpreted geometrically as

- hinge core
- reciprocal shell band
- twelve-phase replication
- toroidal phase carrier

This produces a crystal-like structure where shell states emerge from the hinge and propagate around the phase ring.

12 Parameter Deformation

Along the diagonal family

$$A = B = \lambda,$$

the shell representatives scale approximately as

$$x \sim \lambda, \quad x \sim \frac{1}{\lambda}.$$

Thus the system preserves its reciprocal shell structure under diagonal deformation.

13 Conclusion

The reciprocal rational map studied here exhibits a remarkably simple folded structure.

A self-dual hinge at $x = 1$ generates reciprocal shell classes that can be transported around a toroidal phase carrier.

This framework combines algebraic reciprocity, shell dynamics, and phase topology into a unified geometric interpretation.

Future work includes a deeper investigation of integrability, shell bifurcations, and possible physical realizations of the reciprocal hinge dynamics.

A Derivation of the Lock Equations

B Derivation of the Lock Equations

We derive the folded lock equations used throughout the paper.

B.1 Fixed Locks

Fixed locks satisfy

$$f(x) = x$$

for the reciprocal rational map

$$f(x) = \frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A}.$$

Multiplying both sides gives

$$Ax^4 + Bx^2 + 1 = x(x^4 + Bx^2 + A).$$

Expanding,

$$Ax^4 + Bx^2 + 1 = x^5 + Bx^3 + Ax.$$

Rearranging,

$$x^5 - Ax^4 + Bx^3 - Bx^2 + Ax - 1 = 0.$$

This polynomial is palindromic and can be reduced by introducing

$$z = x^2.$$

Using the reciprocal folding

$$t = x^2 + \frac{1}{x^2},$$

the lock condition reduces to the quadratic

$$t^2 - (A^2 - 2B - 1)t - (A^2 + 2AB - 2A - B^2 - 2B + 1) = 0.$$

B.2 Reciprocal Locks

Reciprocal locks satisfy

$$f(x) = \frac{1}{x}.$$

Substituting the map,

$$\frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A} = \frac{1}{x}.$$

Multiplying gives

$$x(Ax^4 + Bx^2 + 1) = x^4 + Bx^2 + A.$$

Expanding,

$$Ax^5 + Bx^3 + x = x^4 + Bx^2 + A.$$

Rearranging,

$$Ax^5 - x^4 + Bx^3 - Bx^2 + x - A = 0.$$

Using the same reciprocal folding transformation

$$t = x^2 + \frac{1}{x^2},$$

this reduces to

$$A^2t^2 + (A^2 + 2AB - 1)t + (-A^2 + 2AB + 2A + B^2 - 2B - 1) = 0.$$

These two quadratic equations determine the folded shell structure of the reciprocal rational map.